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Abstract

Egypt has faced many grand challenges in recent decades. This project aims to solve some of these challenges, including reducing pollution and improving the use of alternative energies. These challenges were intensified by the reliance on fossil fuels, which contribute significantly to environmental damage and climate change. The purpose of this study is to address the issue of unstable power supply in Abu Simbel City by utilizing the high wind speeds to meet the critical summer electricity demand. The city experiences unreliable energy access due to frequent power system damage caused by high wind speeds. The selected solution for the problem is a hybrid vertical-axis wind turbine that combines Helical Savonius and H-type Darrius blade designs for enhanced efficiency, with gears attached to the central rotating shaft to achieve the required energy output. The prototype underwent extensive testing to verify it fulfilled the design requirements, as it generated about 0.1375 watts in 300 seconds, which is equal to 41.25 watts-second, while maintaining stability and within a volume of 0.35m^3 and a mass of 7 kilograms. This project contributes to solving Egypt's challenges by promoting the adoption of alternative energies and minimizing pollution while improving the quality of life in regions with unstable power supplies.

Introduction

Egypt, like any country, faces some challenges that hinder its development and weaken its economy, agriculture, and industry. One of these challenges is the burning of fossil fuels. Fossil fuel consumption has increased significantly over the last years and reached its peak in 2023 with the consumption of 40,102 TWh of gas, 54,564 TWh of oil, and 45,565 TWh of coal, as shown in **Figure 1**. Burning fossil fuels has various negative consequences, one of which is climate change, as fossil fuel consumption emits greenhouse gases that act as a blanket on the Earth's surface, causing climate change.

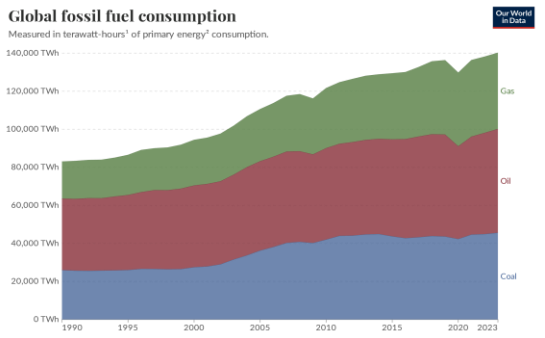


Figure 1: Fossil fuel consumption 1990-2023

The specific problem to be solved is Utilising High Wind Speeds in Abu Simbel to Meet the Critical Summer Electricity Demand, especially in the Abu Simbel area, during the summer. This happens due to the significant increase in electricity consumption in the summer, due to high temperatures and the constant desire to operate cooling devices such as fans and air conditioners. To address such a problem, the Egyptian government resorts to frequent power outages in different areas across the country under the pretext of load shedding.

To solve these problems, several prior solutions were researched to help understand the problem more correctly. One of the prior solutions is El Zafarana wind farm, an Egyptian HAWT located in Egypt, specifically in Ras Gharib. One of its strengths is its location, as El Zafarana specializes in its strong and consistent wind speeds ranging between 8 and to10 m/s, making it an ideal site for wind energy production. One of its weak points is the need for high initial investment Costs. The initial setup of Zafarana required a significant financial investment in turbines, infrastructure, and transmission lines. Another prior solution is Benban Solar Park (Bank, 2023), an Egyptian solar park, which was launched in 2018. One of its strengths is the massive energy production, which reaches up to 3.8 terawatt-hours when operating at full capacity, which is enough for millions of houses. One of its weak points is the high land use. The Benban project used a land area of about 37 square kilometers.

Transitioning to the solution, A VAWT will be made. This VAWT will be a hybrid of helical Savonius and H-Darrieus wind turbines. The helical Savonius will act as the motor that gives the H-Darrieus the external force to work and generate electricity with high efficiency.

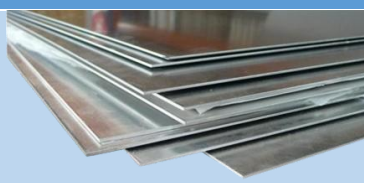
To evaluate the success of the prototype, several prior solutions were set. The ratio between the height of the Savonius to the diameter will be 2:1, and the ratio between the height of the H-Darrieus and the diameter will also be 2:1. The total

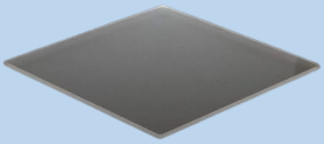
height of the VAWT should be less than 1.5 meters. The swept area should be between 0.8 and 1 m^2. The weight of the prototype should be between 5 and 10 kg. The materials used to implement a solution with such design requirements will be covered in the next section.

Materials & Methods

Table of materials

Table1: Materials used in the implementation of the prototype

Name	Description	Quantity	Image
Aluminum plates	Used to make both the H-Type Darrious and helical Savinous blades	(1m * 3m) ±0.67	
Aluminum Pipe	Used as the central shaft of the VAWT	(1.4m & 1.8m) ±0.67	

Wood	Used to make the box as the base of the VAWT	(0.3m * o.3m * 0.15m) ±0.1	
Dynamo	Convertes the kinetic energy into electric energy	1 dynamo	
Rechargeable batteries	Used to store energy	1 battery	
Ball bearing	Secures the position of the VAWT while rotating	2 bearings	
Acrylic	Used to make gears	0.2m * 0.2m	

Methods

1. 3d Design Development

Created a detailed 3D design using Sketchup, ensuring all dimensions match the requirements of the final design **as shown in figure 2**

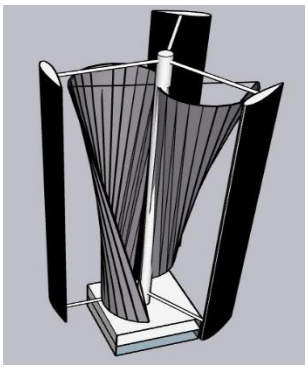


Figure 2: detailed 3D design

2. Blade Fabrication

Transferred the blade dimensions onto the aluminum sheets and precisely cut the sheets **shown in figure 3.**



Figure 3: The cut sheets

3. Formation of Darrius blades

Shaped the cut aluminum sections into airfoil profiles matching the desired measurements of the NACA 0018 air foil and forming them into Darrious blades.



Figure 4 - the Darrius blades had been formed

4. Formation of Savonius blades

Utilised the remaining aluminium material to roll and form the savinous blades using a rolling machine.



Figure 5: The Savonius blade

5. Construction of wooden base

Assembled the wooden components together to form the wooden box to be used as a base to secure the VAWT and contain the circuit and the gear mechanism.



Figure 6: The wooden base

6. Integration of blades with the vertical shaft

- Attached both Savinous and Darrious blades to the vertical shaft, performing welding operations to ensure connections.



Figure 7: The shape of the blades connected to the vertical shaft

7. Assembly of rechargeable circuits

- Integrated the dynamo, TP 4056 module, and rechargeable battery with corresponding gears to enable battery charging. **As shown in figure 8.**



Figure 8: the full recharging circuit

8. **Final assembly of the prototype**

- Connected all fabricated components to complete the prototype, preparing it for the extensive testing.



Figure 9: Final prototype

Test plan

Generated power

It was measured by a multimeter by identifying the generated voltage and the generated amperes then multiplying them together. A graph was created from the data that was collected.

The volume of the prototype

The volume of the prototype was calculated by identifying the area of the H-Darrieus wind turbine and multiplying it by the height of the blade. The maximum volume of the whole prototype should not exceed 0.5 m^3.

The weight of the prototype

The weight of the prototype is an essential part of achieving the design requirements. The maximum weight of the whole prototype should not exceed 10 Kg. After checking its weight using the balance. Then check if it achieves the design requirements.

Results

Negative result

During the construction process, an error occurred in the measurement of the blade attachment points on the central shaft. As a result, incorrect holes were drilled into the shaft, which may affect the alignment and stability of the Savonius blades. To correct this issue, new holes were subsequently drilled using accurate measurements to ensure proper positioning and secure mounting of the blades on the central shaft.

In addition, during the testing phase, the prototype was exposed to high wind speeds, which caused the bearings to be displaced from their original position. As a result, the VAWT stopped operating.

Positive results

The weight of the prototype was measured using a balance to check if the weight of the prototype would exceed 10 kilograms, and it was 7.0 kilograms. As shown in Figure 10.



Figure 10: measuring the weight of the prototype

In addition, the volume of the prototype doesn't exceed 0.5 m^3 And it was 0.35 m^3.

The project succeeded in achieving the required power generation as shown in Table 2, as it generated about 0.1375 watts in 300 seconds, which is equal to 41.25watt watts-second.

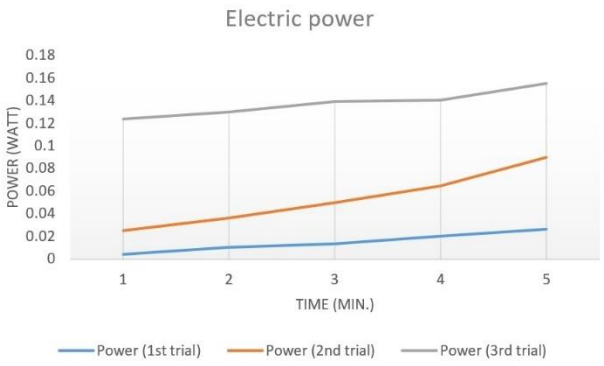


Figure 11: The power generated in the three trials

Table2: generated electricity by the VAWT

Trials	Time	Intensity	Voltage	Power	Average wind speed
1st trial	1 min.	0.0017 A ± 0.17 mA	2.4 V ± 0.012 V	0.0041 W ± 0.046 mW	3.5 m/s
	2 min.	0.0029 A ± 0.29 mA	3.5 V ± 0.0175 V	0.0102 W ± 0.114 mW	

	3 min.	0.0033 A ± 0.33 mA	4.0 V ± 0.02 V	0.0132 W ± 0.148 mW	
	4 min.	0.0046 A ± 0.46 mA	4.3 V ± 0.0215 V	0.0198 W ± 0.221 mW	
	5 min.	0.0054 A ± 0.54 mA	4.88 V ± 0.0244 V	0.0264 W ± 0.295 mW	
average		**0.00358 A ± 0.358 mA**	**3.816 V ± 0.019 V**	**0.0147 W ± 0.164 mW**	
2nd trial	1 min.	0.0068 A ± 0.68 mA	3.7 V ± 0.0185 V	0.0252 W ± 0.282 mW	3.9 m/s
	2 min.	0.0084 A ± 0.84 mA	4.3 V ± 0.0215 V	0.0361 W ± 0.404 mW	
	3 min.	0.010 A ± 1 mA	4.96 V ± 0.0248 V	0.0496 W ± 0.555 mW	
	4 min.	0.013 A ± 13 mA	4.97 V ± 0.012 V	0.0646 W ± 0.722 mW	
	5 min.	0.018 A ± 18 mA	5 V ± 0.025 V	0.09 W ± 1 mW	
average		**0.0112 A ± 112 mA**	**4.586 V ± 0.012 V**	**0.0531 W ± 0.594 mW**	
3rd trial	1 min.	0.025 A ± 25 mA	4.94 V ± 0.0247 V	0.1235 W ± 1.38 mW	4.4 m/s
	2 min.	0.026 A ± 26 mA	5 V ± 0.025 V	0.13 W ± 1.45 mW	
	3 min.	0.028 A ± 28 mA	4.96 V ± 0.0248 V	0.1389 W ± 1.55 mW	
	4 min.	0.028 A ± 28 mA	5 V ± 0.025 V	0.14 W ± 1.57 mW	

	5 min.	0.031 A ± 31 mA	5 V ± 0.025 V	0.155 W ± 1.73 mW	
average		**0.0276 A ± 276 mA**	**4.98 V ± 0.0249 V**	**0.1375 W ± 1.54 mW**	

Analysis

Estimating the gear ratio:

The gear ratio is the ratio between the number of teeth of the driven gear (the gear attached to the dynamo) and the number of teeth of the driver gear (the gear attached to the central shaft of the VAWT). It's also the ratio between the RPM of the driver gear and the RPM of the driven gear (Cobbs & Cranor-Buck, 2011). Therefore, it's an essential factor that must be considered to obtain the required electrical power from the generator. To estimate the required gear ratio, three things should be calculated: the necessary amount of power from the generator, the RPM of the generator that will generate this amount of power, and the RPM of the VAWT.

Firstly, electric power measures the rate of electrical energy transfer by an electric circuit per unit of time. It's measured using the SI unit of power, which is watts or one joule per second. **Equation 1** shows the relation between power, energy, and time (Serway & Vuille, 2011). By substituting with the given energy and time required in the challenge, the amount of required power will be 0.133 W.

$$P = \frac{E}{t}$$

Equation 1: Electric power

Secondly, the RPM of the generator is the measure of the speed at which the generator operates. It's measured, **as shown in equation 2**, by dividing the number of rotations by the time taken. Also, its SI unit is Hertz, but when measuring the speed of engines and generators, RPM is the commonly used unit (Serway & Vuille, 2011). To be able to estimate the RPM of the generator required to get 0.133 W, we need to make a ratio (k_W) between the power generated by the generator at a specific RPM and that RPM using its specifications, **as shown in equation 3**. For example, the generator produces 0.55 W at 6000 RPM. So, the ratio here is about 9.167×10^{-5} W/RPM. Eventually, the required RPM from the generator is about 1454.5 RPM.

$$RPM = \frac{number\ of\ rotations}{time\ in\ minutes}$$

Equation 2: RPM in terms of number of rotations and time

$$RPM_{required} = \frac{P_{required}}{k_W}$$

Equation 3: RPM in terms of power required and the ratio

Thirdly, the Tip Speed Ratio (TSR) of a VAWT is the ratio between the speed of the tips of the wind turbine blades and the wind speed. It's a very important factor that indicates the performance of the VAWT to rotate fast compared to the speed of the wind. **Equation 4** shows how to calculate the TSR in terms of the radius of the VAWT (R) in meters, its rotational speed (ω) in rad/s, and the velocity of the wind in m/s. The average TSR of our hybrid VAWT is about 2.5. So, by substituting in the equation, the rotational speed of the VAWT is about 16.67 rad/s. Then, if we converted this rotational speed to RPM, it would be equal to 159 RPM (Premkumar et al, 2018).

$$TSR = \frac{R \times \omega}{V}$$

Equation 4: TSR of a VAWT

Finally, the required gear ratio (g) can be calculated by dividing the RPM of the driver gear (VAWT's RPM) by the RPM of the driven gear (generator's RPM), **as in Equation 5**, which is equal to 1/9. This means that for every rotation made by the driver gear, the driven gear will rotate 9 times (Cobbs & Cranor-Buck, 2011). **Figure 12 shows** a 3D design for our gear system from a top view.

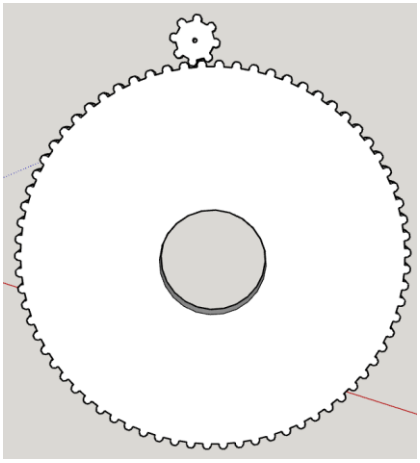


Figure 12: A top view of the gear system

$$g = \frac{RPM_{driver}}{RPM_{driven}}$$

Equation 5: gear ratio

Lift force:

Lift force is an aerodynamic force that acts perpendicular to the wind direction, created when air flows over a curved blade or airfoil. Due to the blade's shape, air moves faster over the top and slower underneath, creating a pressure difference that "lifts" the blade. We can calculate the lift force theoretically by **equation 6**.

$$F_{lift} = \frac{1}{2}\,\rho C_l A v^2$$

Equation 6: Lift Force

In **Equation 6** (Benson, n.d.), $\boldsymbol{F_{lift}}$ is the lift force that will be applied on the H-Darrieus blades, $\boldsymbol{\rho}$ is the density of the air, $\boldsymbol{C_l}$ is the lift coefficient, which is a dimensionless value that represents how effectively the blade generates lift. $\boldsymbol{A}$ is the surface area of the blade, and $\boldsymbol{v}$ is the speed of the wind. Lift is the primary force that is used in the H-Darrieus, as it is more effective than the drag force. The lift force that acts on the H-Darrieus blades is between -1.384 and 1.384 newtons.

Drag force:

Drag force is a resistive aerodynamic force that acts parallel and opposite to the direction of airflow. It occurs due to the friction between the air and the surface area, which slows down the motion of the wind turbine blades. There are two main types of Drag: form drag and skin friction drag. The form drag is caused by the shape of the blade and the difference in the pressure above the upper surface and beneath the lower surface of the blade, while skin friction drag results from air rubbing against the surface of the blade. In the Savonius design, the wind pushes directly against the curved surfaces of the blades, creating torque that spins the rotor. The drag force is calculated using **equation 7**.

$$F_{drag} = \frac{1}{2}\rho C_d A v^2$$

Equation 7: Drag Force

In **equation 7** (Benson, n.d.), $\boldsymbol{F_{drag}}$ is the drag force that will be applied to the helical Savonius blades, $\boldsymbol{\rho}$ is the density of the air, $\boldsymbol{C_d}$ is the drag coefficient, which is a dimensionless value that represents how effectively the blade generates drag. $\boldsymbol{A}$ is the surface area of the blade, and $\boldsymbol{v}$ is the speed of the wind. The drag force that acts on the H-Darrieus blades is approximately 1.115.

In a VAWT, lift and drag forces play crucial roles in its operation and efficiency. Lift is primarily responsible for generating torque in lift-based VAWTs, such as the H-Darrieus turbine. This force is created by the difference in pressure on the curved surfaces of the blades, allowing the blades to rotate around the vertical axis. On the other hand, drag is mainly utilized in drag-based VAWTs, like the Savonius turbine. In these designs, the wind pushes directly against the blades, causing rotation.

The aspect ratio (AR)

The aspect ratio (AR) of a hybrid Vertical Axis Wind Turbine (VAWT), defined as:

$$AR = \frac{H}{D}$$

Equation 8: The aspect ratio

(where H is rotor height and D is diameter), The height of the Savonius is equal to 1 m, and the height of the Darrius is equal to 1.2 m, and the diameter of the Savonius is 0.5 m, and the diameter of the Darrius is 0.6 m. Aspect ratio plays a crucial role in determining the turbine's aerodynamic performance and startup behavior. A higher aspect ratio increases the swept area and reduces the moment of inertia:

$$I = \frac{1}{2}mr^2$$

Equation 9: moment of inertia

This lower inertia results in faster angular acceleration, and as the moment of inertia is equal to 0.0337 kg·m^2 as there is an inverse relation between moment of inertia and the angular acceleration.

$$a = \frac{T}{I}$$

Equation 10: angular acceleration

which helps the turbine start more quickly in low or variable wind conditions. Additionally, it improves torque delivery to the generator and overall energy output. However, the high aspect ratio in the Eco Spin requires stronger structural support. In contrast, a lower aspect ratio slightly improves the power coefficient:

$$C_p = \frac{P}{\frac{1}{2}\rho A V^3}$$

Equation 11: power coefficient

but increases the moment of inertia and reduces vertical wind capture. The chosen aspect ratio for Eco Spin is 2 to achieve a lower moment of inertia and, additionally, a higher angular acceleration

Table3: Learning Outcomes

Subject	LO	Concept	Usage
Chemistry	LO.1.09	Texture, Strength, Elasticity, Malleability.	Choosing the prototype's materials according to the required physical properties.
Chemistry	LO.1.14	Electrochemical cells.	Knowing how the battery works and choosing a good rechargeable battery.
Earth Science	LO.1.10	Directly capturing the wind energy through turbines to provide electricity.	Learning about wind turbines and how they work to produce electric energy from environmentally friendly sources.
Mechanics	LO.1.06	Resultant of forces and resolution of a force	Understanding how lift and drag forces affect the wind turbine.
Physics	LO.1.09	Laminar vs Turbulent Flow	Understanding types of wind flow that the prototype will be subjected to

As shown in Table 3, these are the LOs that helped in completing this project using the LOs in the curriculum.

Conclusions

Based on the previous research on Egypt's grand challenges, particularly the severe need to reduce pollution and improve the use of alternative energy, and after addressing the issue of unstable power supply in Abu Simbel City, it was determined that an effective solution to this problem is the development of a hybrid vertical-axis wind turbine. The turbine features a combination of the H-type Darrious and Helical Savonius blades, combined with a gear mechanism, designed to harness high wind speeds and provide stable energy output. Testing the prototype showed promising results, as it generated about 0.1375 watts in 300 seconds, which is equal to 41.25 watts-second, while maintaining stability and within a volume of $0.35m^3$ and a mass

of 7 kilograms. It also showed how the hybrid blades' design performed during rapid wind typical of Abu Simbel. Overall, the project is a significant step towards solving the problem of pollution and ending the reliance on fossil fuels in Egypt, besides solving the power insecurity problem present in Abu Simbel City.

Recommendation

Real-life application:

It is recommended to implement this project in Abu Simbel and nearby areas southwest of Aswan, as shown in **Figure 13**, for two main reasons. First, the region has high wind speeds ranging from 2.1 to 8.8 m/s (El-Mekawy et al., 2018). Between 1996 and 2013, the highest wind frequency occurred in autumn, reaching 84.3%, with wind speeds between 3.6 and 5.7 m/s—levels considered uncomfortable and unusually high (Guo et al., 2015). Despite this, wind energy is not being utilized. Notably, on May 26, 2023, Abu Simbel experienced a full power outage due to damage to a high-voltage tower caused by a dust storm and strong

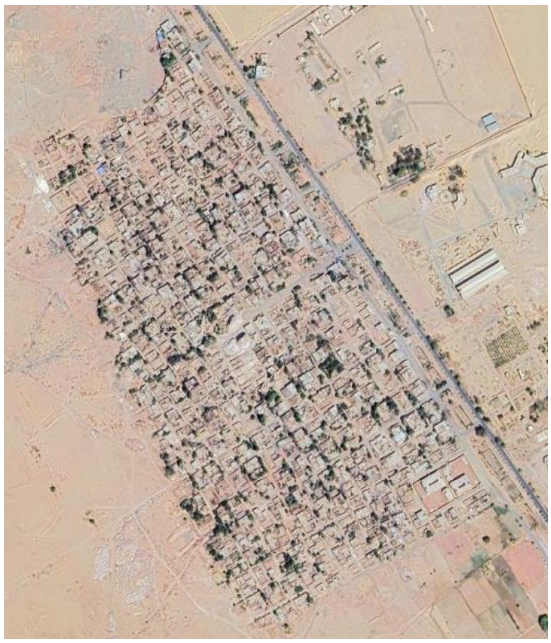


Figure 13: a residential area in the Abu Simbel area, southwest of Aswan

winds (Al-Subaie, 2023). The second reason is the area's extreme heat. Located in Upper Egypt, Abu Simbel sees average temperatures above 24°C, with maximums reaching 42°C (Dave, 2023). This creates high electricity demand for cooling, often resulting in power outages due to fuel shortages and overconsumption.

The project's prototype will be built at a 1:1 scale to serve as a rooftop product for residents. It aims to generate enough electricity to run basic air conditioning devices, such as fans. The hybrid VAWT will have a height of 1.2 m and a diameter of 0.6 m, giving it an H/D ratio of 2:1. The wooden box measures 0.3 × 0.3 × 0.15 m. The Darrieus blades will be NACA 0018 type, with a chord length of approximately 0.16 m.

Using carbon-fiber reinforced plastic

One advanced recommendation for improving our prototype in real life would be to use custom aerospace-composite materials for the turbine blades and other main structural parts, these materials include carbon-fiber reinforced plastics, which would make the VAWT stronger, lighter, and more efficient.

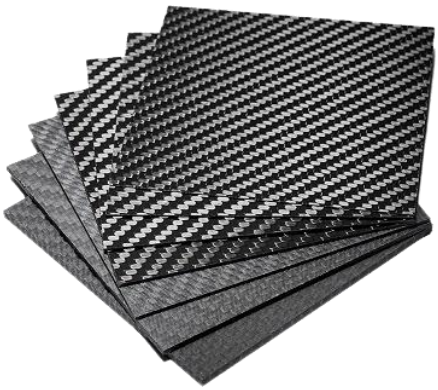


Figure 14: Carbon-fiber reinforced plastic sheets

For example, using carbon-fiber reinforced plastic would make the VAWT's blades much lighter than if the blades were made up of aluminum, which will reduce the

rotational inertia, allowing the turbine to start at lower wind speeds and respond quicker to wind changes. Furthermore, carbon-fiber reinforced plastic has a higher strength-to-weight ratio, which means blades can be made longer and thinner without bending or deforming, allowing for a larger swept area and a better aerodynamic design with longer-lasting blades that maintain their shape and performance over time.

However, This material is very expensive and hard to manufacture, as it requires specialized manufacturing equipment, such as autoclaves and precision molds, which aren't available for the small scale. Working with aerospace composites also requires advanced technical skills and a lot of safety procedures because of the complex manufacturing process. These resources are only found in large companies or advanced research labs.

Implement of a Permanent Magnet Synchronous Generator

To improve the Eco Spin in real life, a bigger generator will be used to achieve the highest efficiency of generating electricity in a VAWT. A permanent magnet synchronous generator (PMSG) creates a magnetic field by

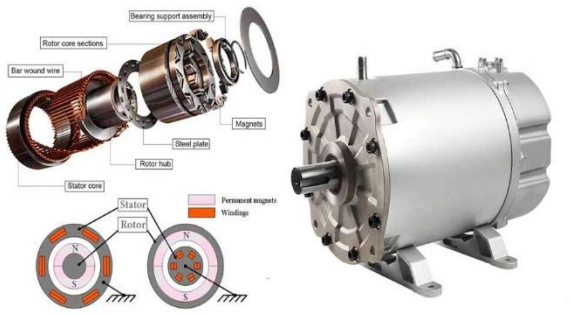


Figure 15: Diagram of the Permanent Magnet Synchronous Generator

rotating its permanent magnet and produces an AC by converting the mechanical energy into electrical energy. The (PMSG) consists of the stator (the outer part), the rotor, and copper wire loops. The rotor spins due to the magnetic field produced by the PMSG, eliminating any need for an external excitation source. After that, the magnetic field lines cut across the stationary stator coils, causing a change in the magnetic field, which, according to Faraday's law that states that any change in the magnetic field induces an electric current. Other advantages of the PMSG include its high-power density and loss reduction.

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